

Rational Functions and Asymptotes



Finding asymptotes and intercepts

- 1 For $f(x) = \frac{x+2}{x-3}$, find:
 - a the vertical asymptote
 - b the horizontal asymptote
 - c the x-intercept
 - d the y-intercept
 - 2 Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.
 - 3 Consider $f(x) = \frac{x^2-9}{x-3}$.
 - a Simplify f and identify the feature at $x = 3$.
 - b Does the graph have a vertical asymptote?
 - 4 Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.
-

Behaviour near asymptotes

- 5 For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:
 - a $x \rightarrow 3^+$
 - b $x \rightarrow 3^-$
 - c $x \rightarrow \infty$
- 6 Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.